

CONVOLUCIÓN-SLIT

Ing Leandro Borgnino

Teoría de Señales y Sistemas Lineales

Guía Prácticos 2

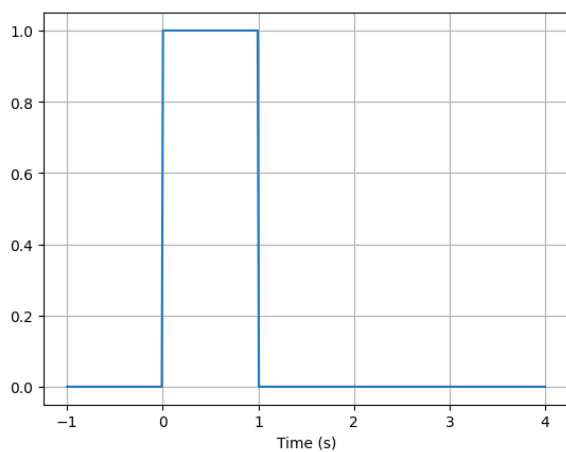
Tema 1: Convolución

(1) Graficar las señales, determinar la convolución entre $x(t)$ y $h(t)$ y graficar el resultado:

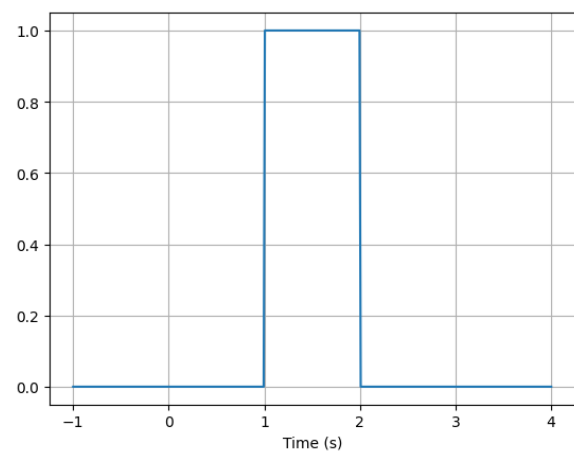
1.

$$h(t) = \begin{cases} 1, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$x(t) = \begin{cases} 1, & 1 \leq t \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$



(a) $x(t)$



(b) $h(t)$

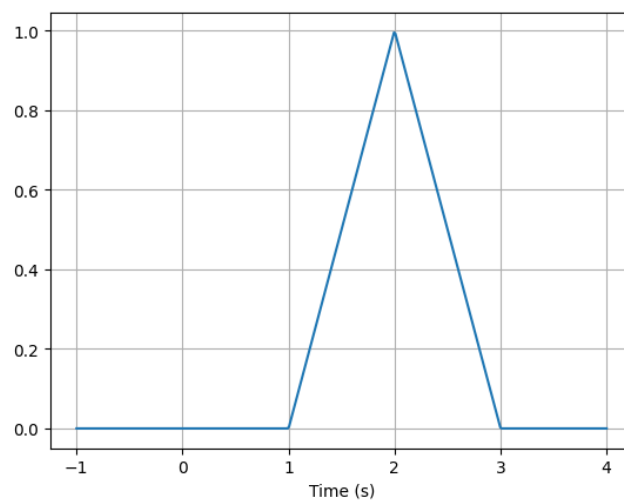


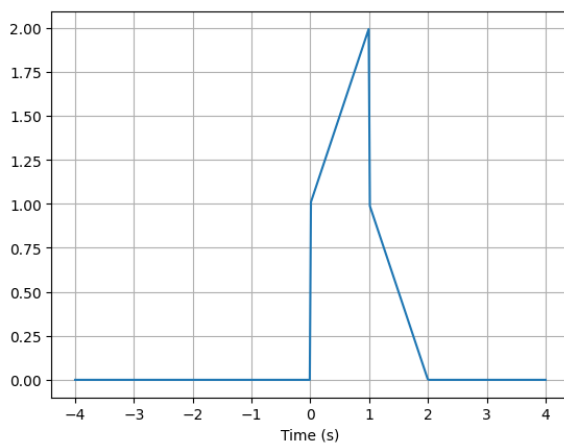
Figure 2: $y(t)$

$$y(t) = \begin{cases} 0, & t < 1 \\ t - 1, & 1 \leq t \leq 2 \\ -t + 3, & 2 \leq t \leq 3 \\ 0, & t > 3 \end{cases}$$

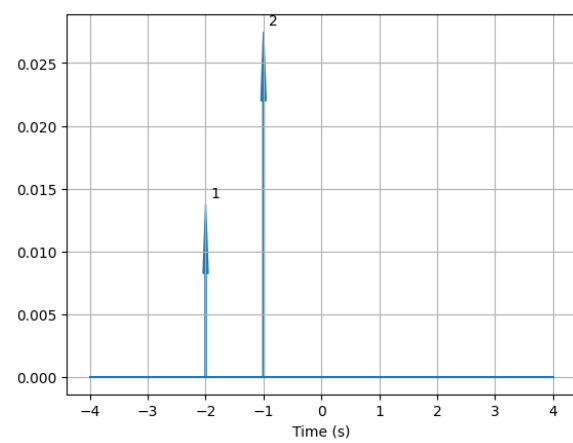
2.

$$x(t) = \begin{cases} t + 1, & 0 \leq t \leq 1 \\ 2 - t, & 1 < t \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

$$h(t) = \delta(t + 2) + 2\delta(t + 1)$$



(a) $x(t)$



(b) $h(t)$

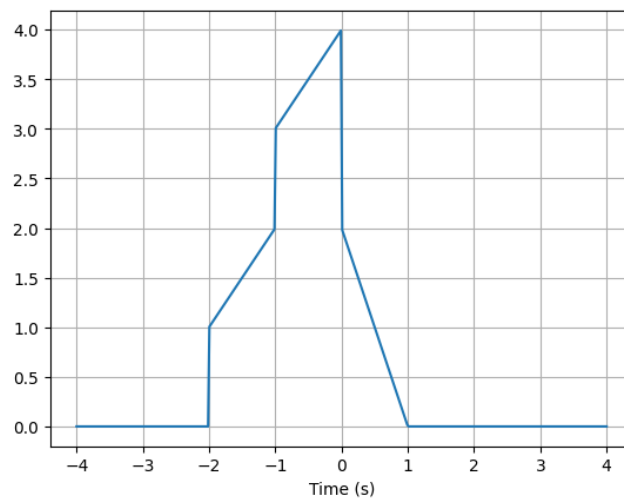


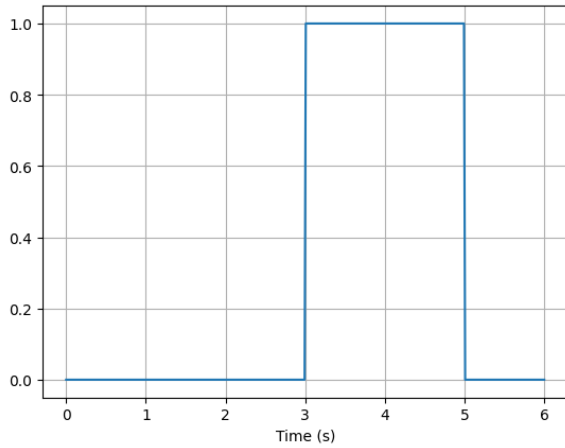
Figure 4: $y(t)$

$$y(t) = \begin{cases} 0 & t < -2 \\ t + 3 & -2 \leq t < -1 \\ t + 4 & -1 \leq t < 0 \\ -2t + 2 & 0 \leq t < 1 \\ 0 & t \geq 1 \end{cases} \quad (1.1)$$

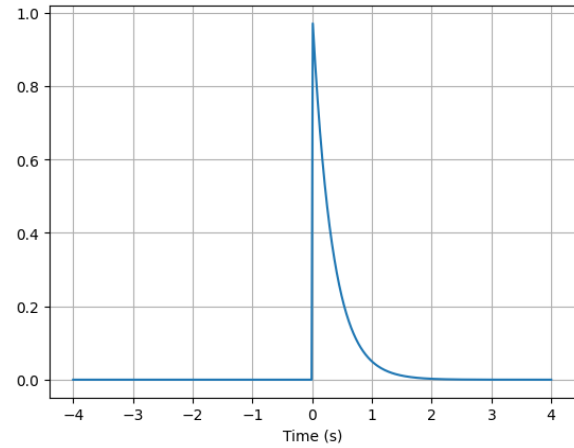
3.

$$x(t) = u(t - 3) - u(t - 5)$$

$$h(t) = e^{-3t}u(t)$$



(a) $x(t)$



(b) $h(t)$

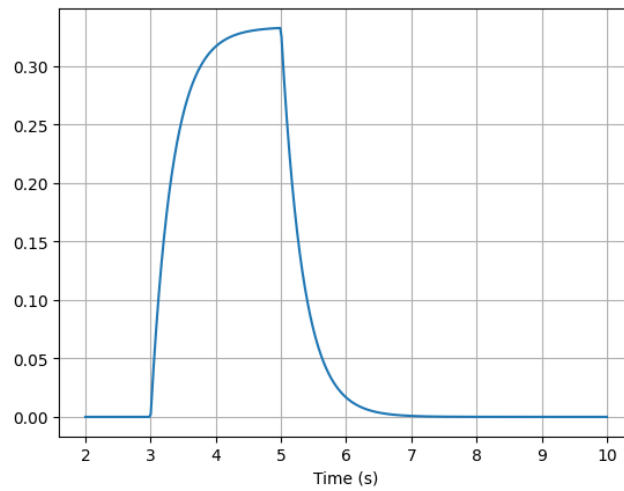


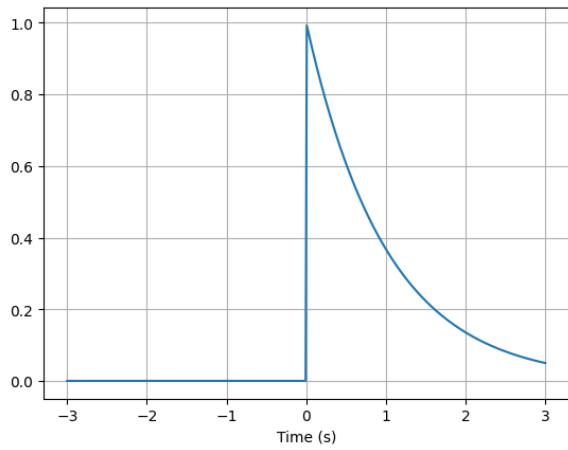
Figure 6: $y(t)$

$$y(t) = \begin{cases} 0 & t < 3 \\ -\frac{1}{3}e^{-3t+9} + \frac{1}{3} & 3 \leq t < 5 \\ -\frac{1}{3}e^{-3t}(e^9 - e^{15}) & 5 \leq t < \infty \end{cases} \quad (1.2)$$

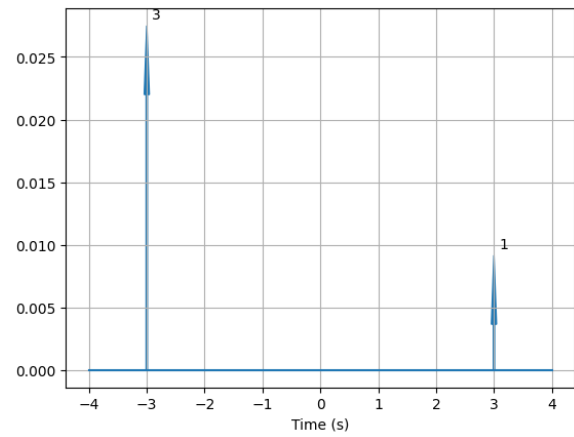
4.

$$x(t) = e^{-t}u(t)$$

$$h(t) = \delta(t - 3) + 3\delta(t + 3)$$



(a) $x(t)$



(b) $h(t)$

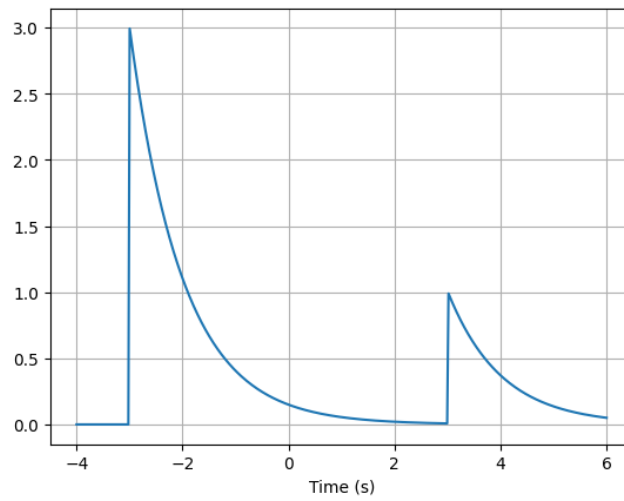


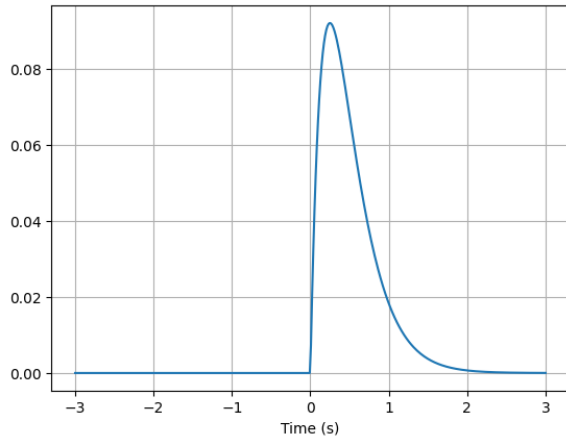
Figure 8: $y(t)$

$$y(t) = 3e^{-t+3} (u(t+3) - u(t-3)) + e^{-t}u(t-3)[3e^{-3} + e^3]$$

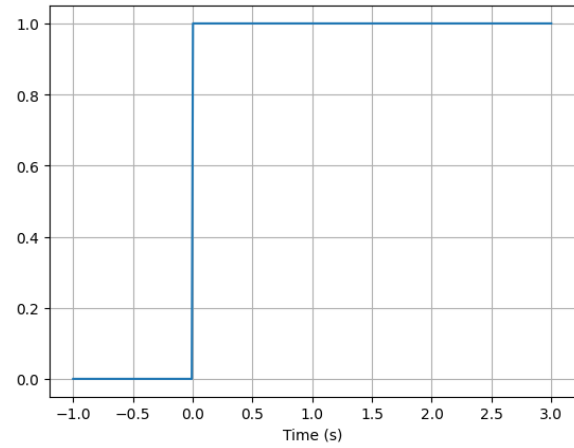
5.

$$x(t) = te^{-\alpha t}u(t)$$

$$h(t) = u(t)$$



(a) $x(t)$



(b) $h(t)$

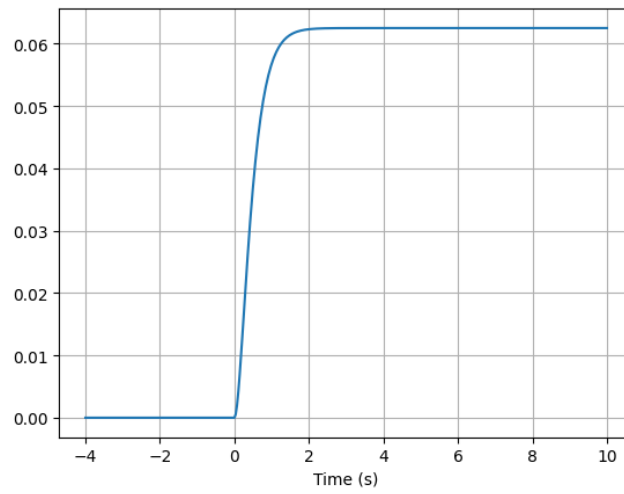


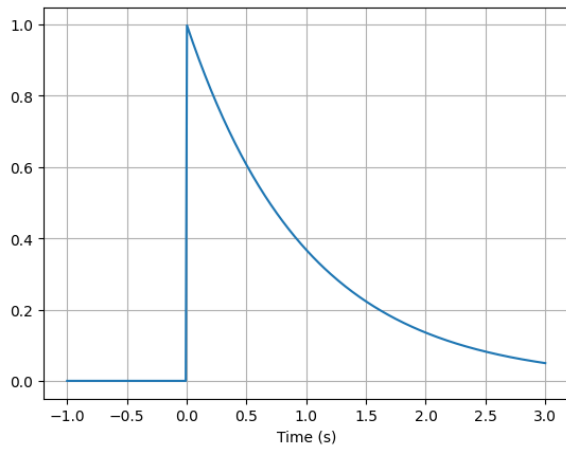
Figure 10: $y(t)$

$$y(t) = \frac{1 - e^{-\alpha t}(\alpha t + 1)}{\alpha^2}$$

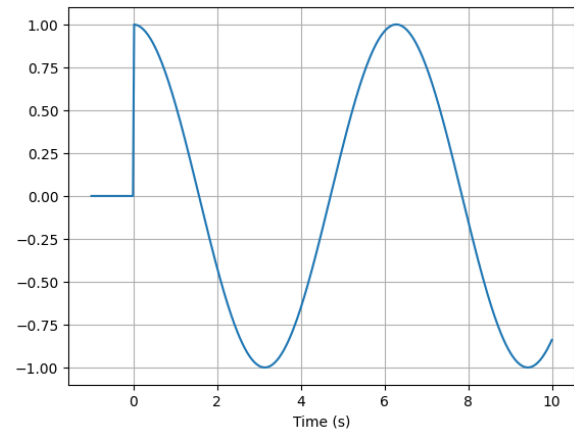
6.

$$x(t) = e^{-t}u(t)$$

$$h(t) = \cos(t)u(t)$$



(a) $x(t)$



(b) $h(t)$

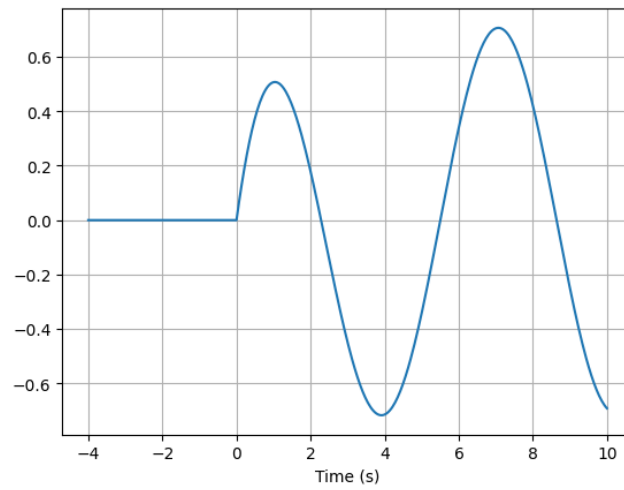


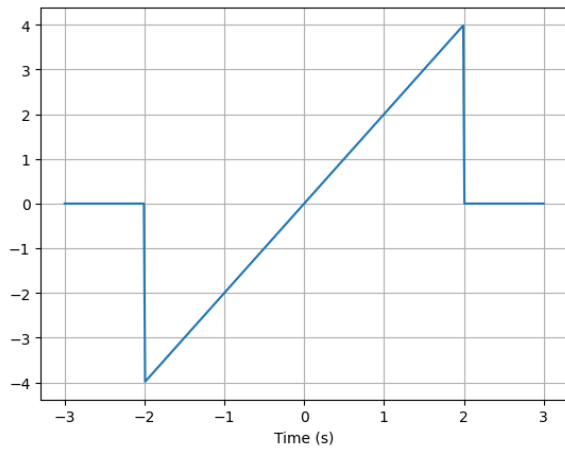
Figure 12: $y(t)$

$$y(t) = -\frac{1}{2}e^{-t} + \frac{1}{2}(\sin(t) + \cos(t))$$

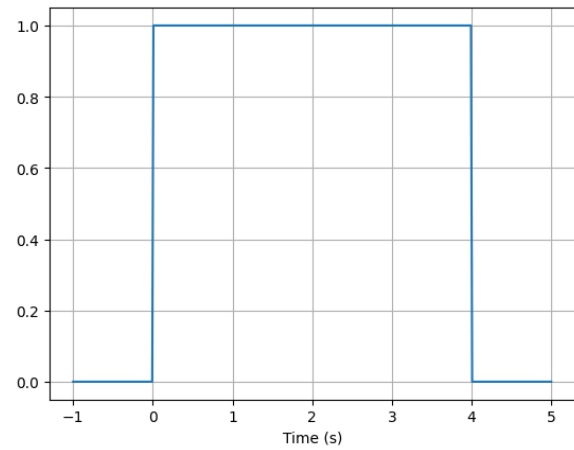
7.

$$x(t) = 2t[u(t+2) - u(t-2)]$$

$$h(t) = u(t) - u(t-4)$$



(a) $x(t)$



(b) $h(t)$

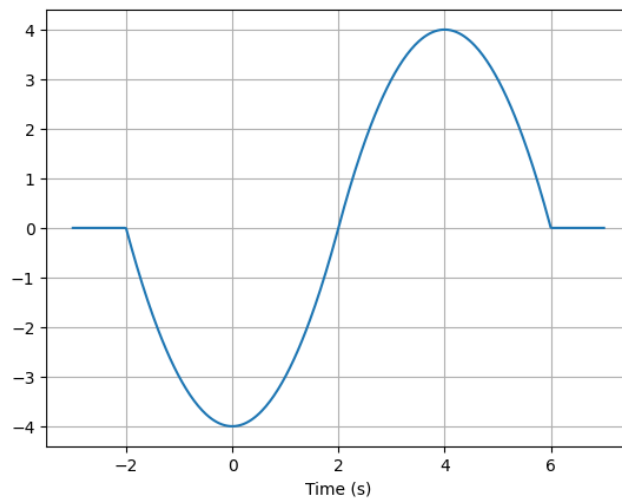
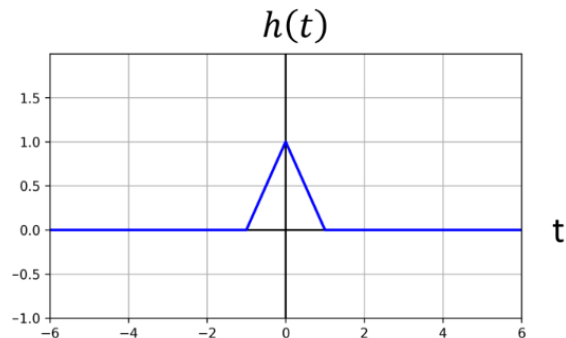
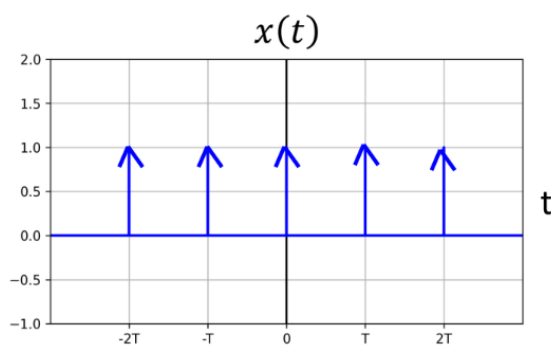


Figure 14: $y(t)$

$$y(t) = \begin{cases} 0 & t < -2 \\ t^2 - 4 & -2 \leq t < 2 \\ -t^2 + 8t - 12 & 2 \leq t < 6 \\ 0 & t \geq 6 \end{cases} \quad (1.3)$$

(2) Expresar las señales gráficas de forma matemática, determinar la convolución entre $x(t)$ y $h(t)$ y graficar el resultado:



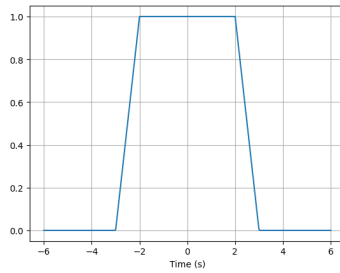
1. Señales a analizar:

$$x(t) = \sum_{n=-2}^2 \delta(t - nT) \quad (1.4)$$

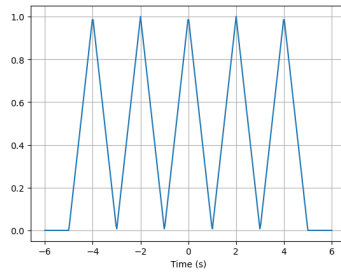
$$h(t) = (t + 1)(u(t + 1) - u(t)) + (-t + 1)(u(t) - u(t - 1)) \quad (1.5)$$

$$y(t) = \sum_{n=-2}^2 h(t - nT) = \sum_{n=-2}^2 (t - nT + 1)(u(t - nT + 1) - u(t - nT)) + (-t + nT + 1)(u(t - nT) - u(t - nT - 1)) \quad (1.6)$$

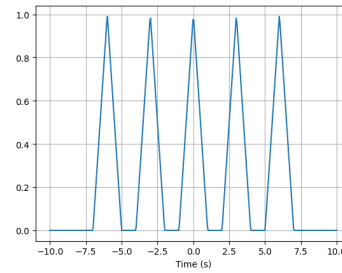
Por ejemplo:



(a) $y(t)$ para $T = 1$

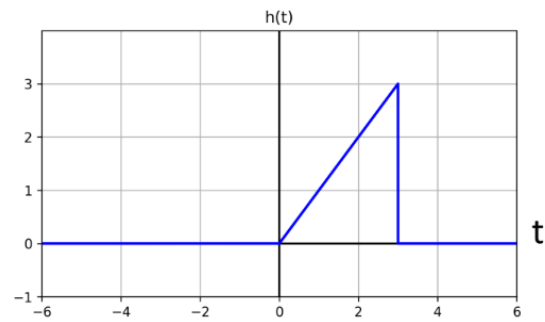
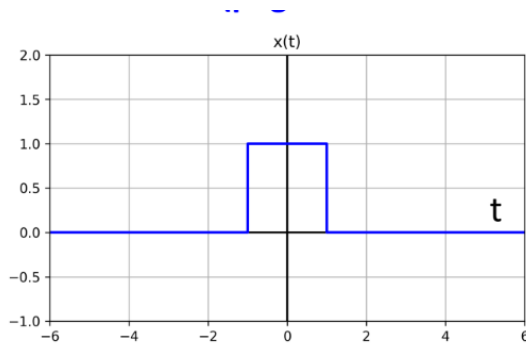


(b) $y(t)$ para $T = 2$



(c) $y(t)$ para $T = 3$

2. Señales a analizar:



$$x(t) = u(t + 1) - u(t - 1)$$

$$h(t) = t * (u(t) - u(t - 3))$$

$$y(t) = \begin{cases} 0 & t < -1 \\ \frac{t^2}{2} + t + \frac{1}{2} & -1 \leq t < 1 \\ 2t & 1 \leq t < 2 \\ -\frac{t^2}{2} + t + 4 & 2 \leq t < 4 \\ 0 & t \geq 4 \end{cases} \quad (1.7)$$

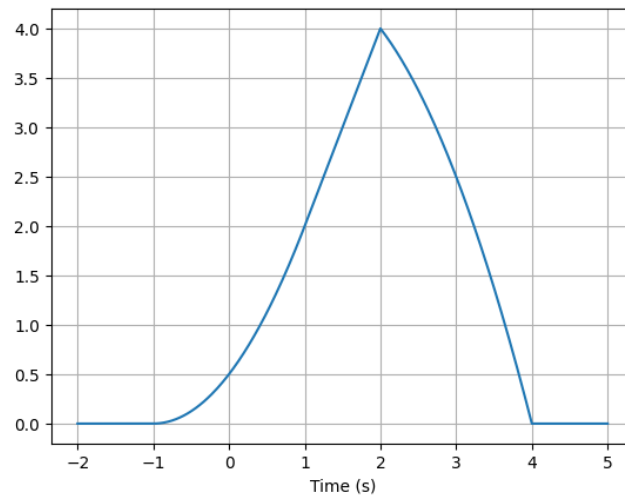
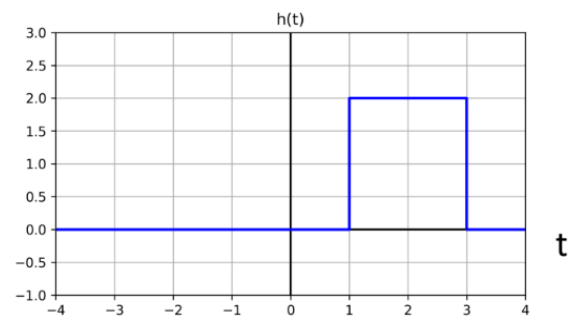
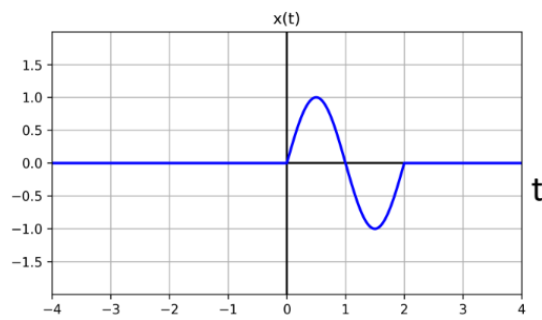


Figure 16: $y(t)$

3. Señales a analizar:



$$x(t) = \sin(\pi t)[u(t) - u(t - 2)]$$

$$h(t) = 2[u(t - 1) - u(t - 3)]$$

$$y(t) = \begin{cases} 0 & t < 1 \\ -\frac{2}{\pi}[\cos(\pi(t - 1)) - 1] & 1 \leq t < 3 \\ \frac{2}{\pi}[\cos(\pi(t - 3)) - 1] & 3 \leq t < 5 \\ 0 & t \geq 5 \end{cases} \quad (1.8)$$

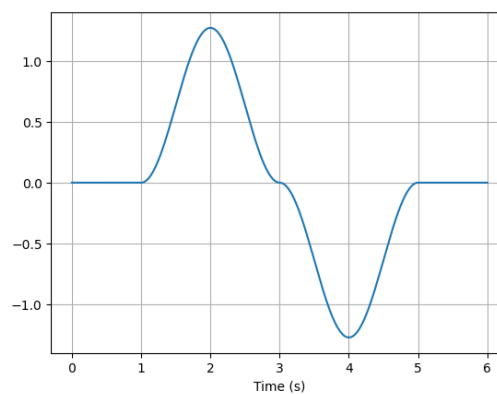
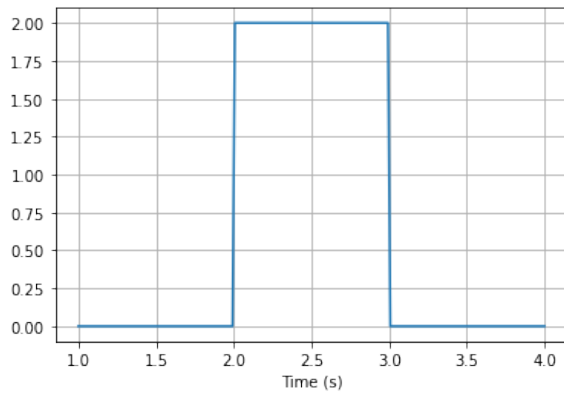
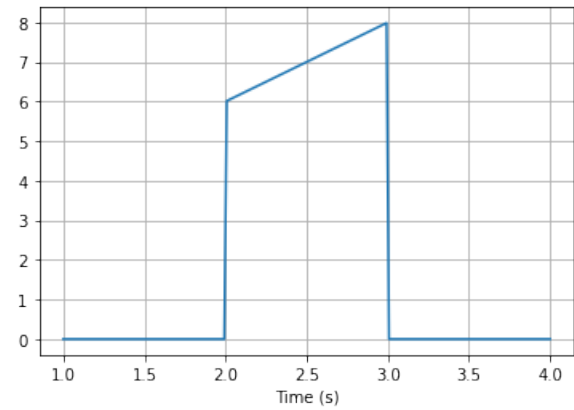


Figure 17: $y(t)$

4. Señales a analizar:



(a) $x(t)$



(b) $h(t)$

$$x(t) = 2[u(t-2) - u(t-3)]$$

$$h(t) = (2t+2)(u(t-2) - u(t-3))$$

$$y(t) = \begin{cases} 0 & t < 4 \\ 2t^2 - 4t - 16 & 4 \leq t < 5 \\ -2t^2 + 8t + 24 & 5 \leq t < 6 \\ 0 & t \geq 6 \end{cases} \quad (1.9)$$

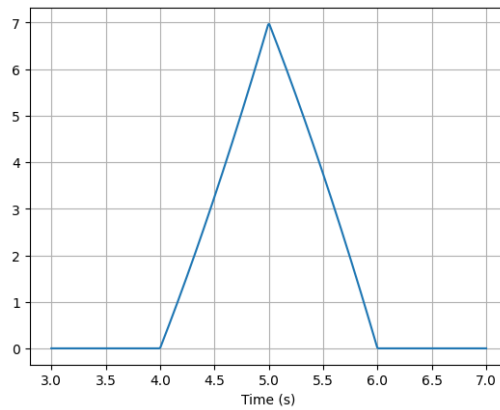
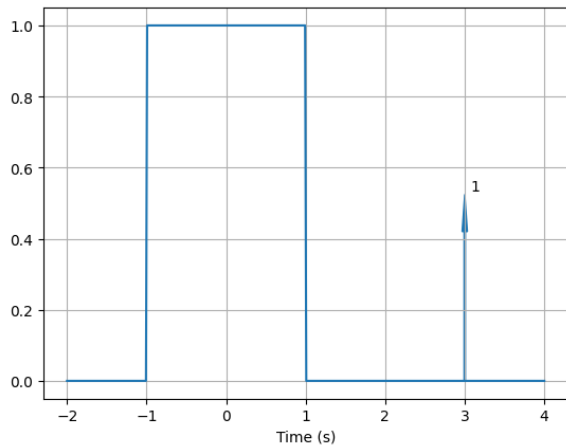
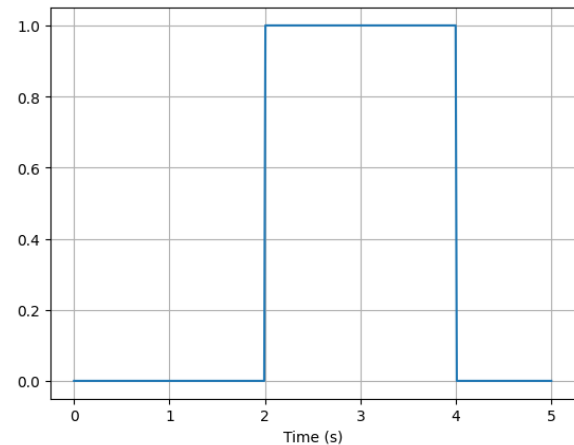


Figure 19: $y(t)$

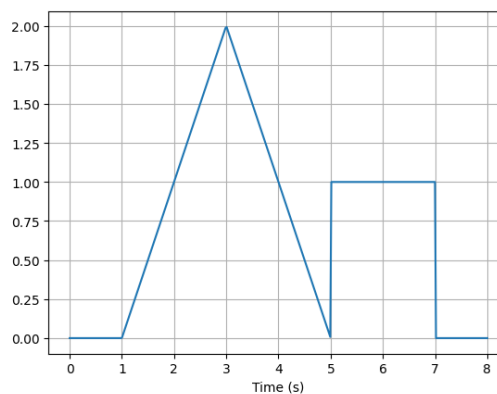
5. Señales a analizar:


 (a) $x(t)$

 (b) $h(t)$

$$x(t) = u(t+1) - u(t-1) + \delta(t-3)$$

$$h(t) = u(t-2) - u(t-4)$$

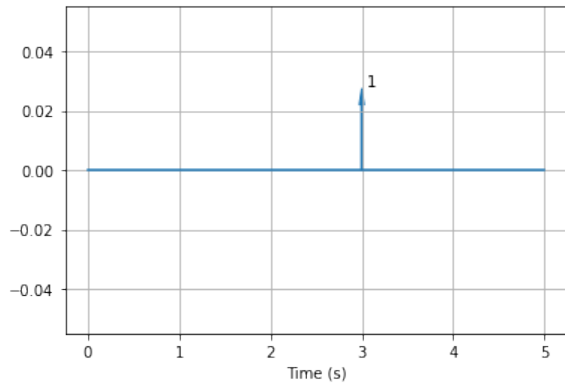
$$y(t) = \begin{cases} 0 & t < 1 \\ t-3 & 1 \leq t < 3 \\ -t+5 & 3 \leq t < 5 \\ 1 & 5 \leq t < 7 \\ 0 & t \geq 7 \end{cases} \quad (1.10)$$


 Figure 21: $y(t)$

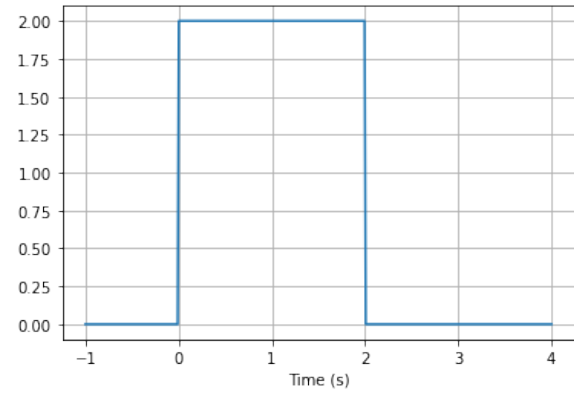
Tema 2: Características de los sistemas según su respuesta al impulso

(1) Describir en cada sistema, según su respuesta al impulso $h(t)$, si tiene memoria, si es estable, si es causal. Además determinar la respuesta del sistema para la señal $x(t)$ y expresarlo de forma matemática y gráfica.

1. Señales a analizar:



(a) $x_1(t)$



(b) $h(t)$

$$x(t) = \delta(t - 3)$$

$$h(t) = u(t) - u(t - 2)$$

El sistema tiene memoria, es estable y es causal.

$$y(t) = u(t - 3) - u(t - 5)$$

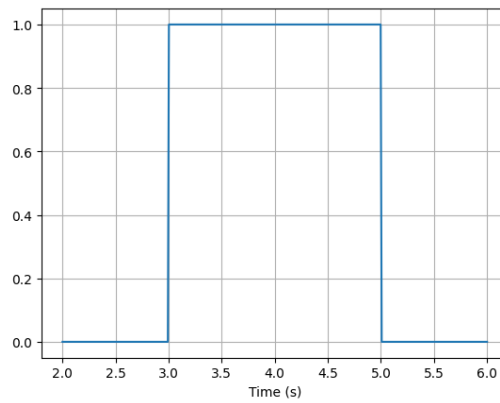


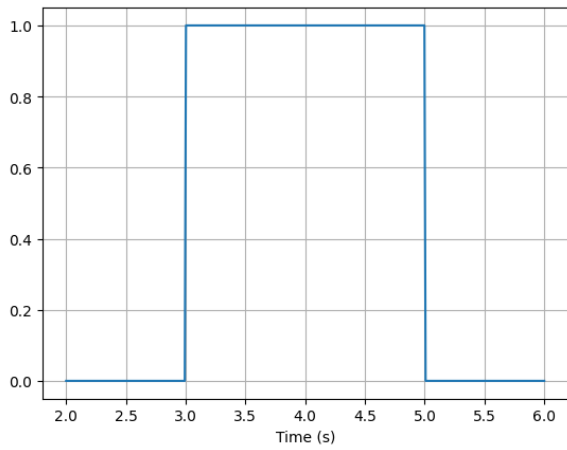
Figure 23: $y(t)$

2. Señales a analizar:

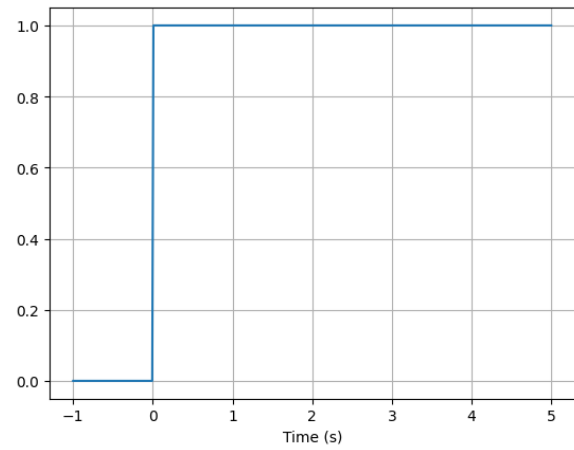
$$x(t) = u(t - 3) - u(t - 5)$$

$$h(t) = u(t)$$

El sistema tiene memoria, no es estable y es causal.



(a) $x(t)$



(b) $h(t)$

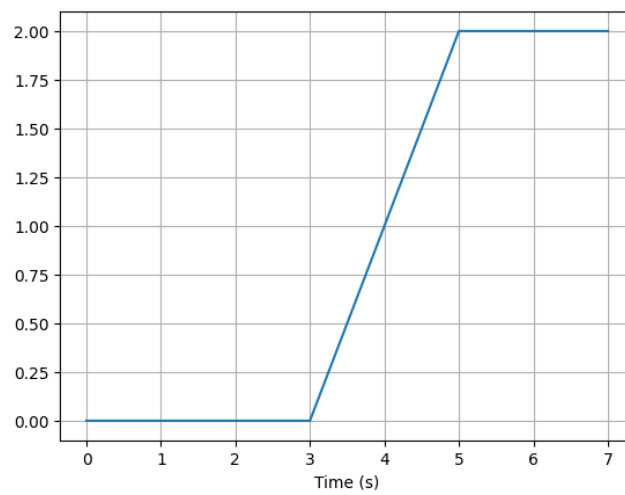
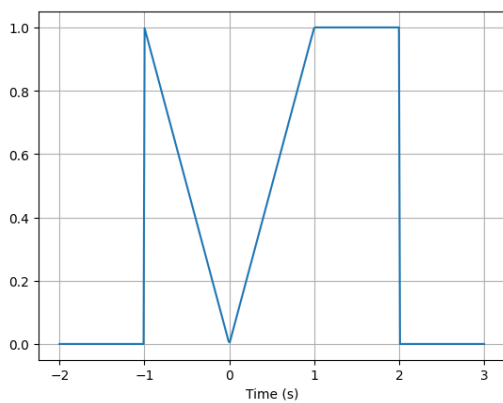


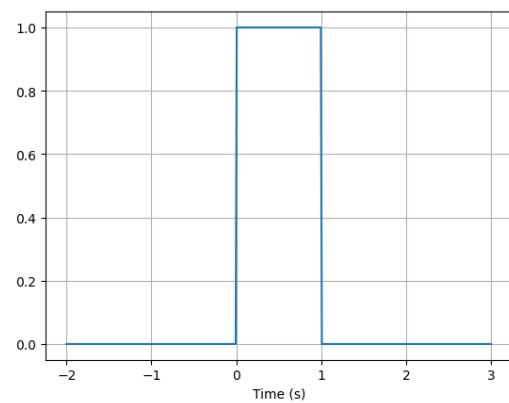
Figure 25: $y(t)$

$$y(t) = \begin{cases} 0 & t < 3 \\ t - 3 & 3 \leq t < 5 \\ 2 & t \geq 5 \end{cases} \quad (2.1)$$

3. Señales a analizar:



(a) $x(t)$



(b) $h(t)$

$$x(t) = -t(u(t+1) - u(t)) + t(u(t) - u(t-1)) + u(t-1) - u(t-2)$$

$$h(t) = u(t) - u(t-1)$$

El sistema tiene memoria, es estable y causal.

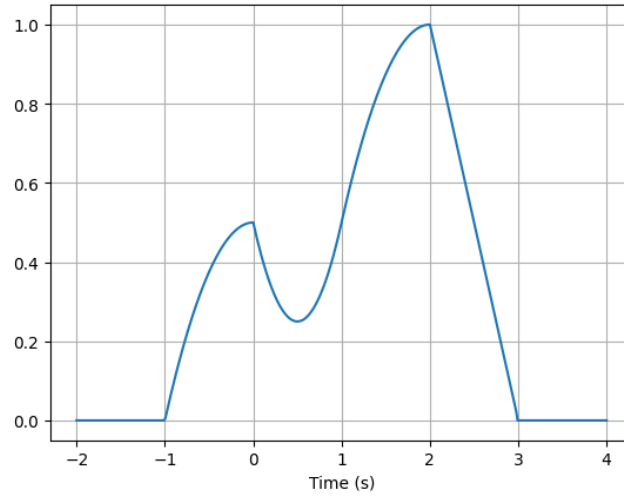


Figure 27: $y(t)$

$$y(t) = \begin{cases} 0 & t < -1 \\ -\frac{t^2}{2} + \frac{1}{2} & -1 \leq t < 0 \\ t^2 - t + \frac{1}{2} & 0 \leq t < 1 \\ -\frac{t^2}{2} + 2t - 1 & 1 \leq t < 2 \\ -t + 3 & 2 \leq t < 3 \\ 0 & t \geq 3 \end{cases} \quad (2.2)$$